

Every amenable action admits a one-approximating sequence

Candidate full solution to the conjecture after Definition 1.3 of arXiv:1611.08486

Autonomous proof packet — agent_lane_09

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Status. Candidate full solution, likely valid, pending expert review. The proof covers locally compact Hausdorff groups in the net convention used by the source paper. No unimodularity assumption is made.

1 Source question

The source is A. M. González-Pérez, *Crossed-Product Extensions of L_p -Bounds for Amenable Actions*, arXiv:1611.08486, published in *J. Functional Analysis* 274 (2018), 2846–2883 [1]. On page 9, after Definition 1.3, it asks whether every amenable action has a C -approximating sequence with $C = 1$.

supp $[\iota_*\nu] \subset E$. Restricting to such support amounts to removing a null set of X , see [AEG94 Lemma 1.3]. Similar results follow for ν -preserving actions when ν is an infinite regular measure changing closed sets by locally closed sets.

As a corollary of the following discussion we obtain that any action $\theta : G \rightarrow \text{Aut}(\mathcal{M})$ of G on a von Neumann algebra $\mathcal{M}$ is amenable iff it has an approximating sequence $(x_\alpha)_\alpha \subset C_c(G; \mathcal{Z}(\mathcal{M}))$ as in Lemma 1.2. We introduce now the refinement of amenability of actions that we are going to use through the next subsections.

Definition 1.3. Let $(\mathcal{M}, \tau)$ be a semifinite von Neumann algebra and denote $(\mathcal{Z}(\mathcal{M}), \tau|_{\mathcal{Z}(\mathcal{M})})$ by $(L_\infty(X), \nu)$. We say that the action $\theta : G \rightarrow \text{Aut}(\mathcal{M})$ has a C -approximating sequence iff there is a sequence of sets $F_\alpha \subset X \times G$ such that

$$1 \leq \text{ess sup}_x \mu\{g \in G : (x, g) \in F_\alpha\} \leq C \text{ess inf}_x \mu\{g \in G : (x, g) \in F_\alpha\} < \infty, \quad (1.3)$$

and the elements

$$x_\alpha(x, g) = \frac{\chi_{F_\alpha}(x, g)}{\mu\{g \in G : (x, g) \in F_\alpha\}^{\frac{1}{2}}}$$

form an approximating sequence satisfying (iii) in Lemma 1.2

Many natural amenable actions, for example that of $\mathbb{F}_r$ in its hyperbolic boundary, admit a 1-approximating sequence. For instance, we can just take F_m the set of pairs $(\omega, \eta) \in \partial\mathbb{F}_r \times \mathbb{F}_r$ such that ω is an infinite reduced word and η is one of the m initial subwords of length less than m . The existence of C -approximating sequences is stable under natural operations like tensor product extensions $\text{Id} \otimes \theta : G \rightarrow \text{Aut}(\mathcal{M} \overline{\otimes} \mathcal{M}_2)$, diagonal products $\theta_1 \times \theta_2 : G \rightarrow \text{Aut}(\mathcal{M}_1 \overline{\otimes} \mathcal{M}_2)$ or tensor products $\theta_1 \otimes \theta_2 : G_1 \times G_2 \rightarrow \text{Aut}(\mathcal{M}_1 \overline{\otimes} \mathcal{M}_2)$. We conjecture that every amenable action admits C -approximating sequences with $C = 1$.

Figure 1: Definition 1.3 and the conjecture on source-paper page 9.

Write the centre of the semifinite von Neumann algebra as $\mathcal{Z}(\mathcal{M}) = L_\infty(X)$. For a measurable $F \subset X \times G$, put

$$F_x = \{h \in G : (x, h) \in F\}.$$

The case $C = 1$ means that $\mu(F_x)$ is independent of x (up to null sets), and the normalized indicators

$$z_F(x, h) = \mu(F_x)^{-1/2} \mathbf{1}_{F_x}(h)$$

must be asymptotically equivariant. When all fibres have measure L , the squared fibrewise L_2 defect is exactly

$$\int_G |z_F(g^{-1}x, g^{-1}h) - z_F(x, h)|^2 d\mu(h) = \frac{\mu(gF_{g^{-1}x} \Delta F_x)}{L}. \quad (1)$$

2 Proof intuition

Amenability supplies positive compactly supported fibre functions u_x of L_2 norm one. Their squares $p_x = u_x^2$ are probability densities with small L_1 equivariance defect. The subgraph of p_x would be an equal-volume set, but it lives in $G \times \mathbb{R}_+$ rather than in G . We replace the vertical coordinate by finitely many *right-translation slots*: quantized level sets of p_x are placed in mutually disjoint sets $A_j(x)r_j \subset G$. Left translation commutes with these right slots, so the layer-cake estimate controls their symmetric-difference defect.

The finite layer sum misses its target volume by a bounded amount depending on x . One extra slot supplies exactly that missing measure. This correction can behave arbitrarily under the action: its measure is uniformly bounded, whereas the total fibre measure tends to infinity, so its relative defect tends to zero.

Right translation need not preserve left Haar measure. This is overcome by placing every slot in $H = \ker \Delta_G$. If G is nonamenable, then H is noncompact: otherwise H would be amenable and the abelian quotient G/H would make G an amenable extension. Thus H supplies arbitrarily many disjoint Haar-preserving slots. If G itself is amenable, ordinary Følner sets already solve the problem for every action.

3 Statement

We use the source paper's characterization of amenability: there is a net of positive $u_\alpha \in C_c(G; \mathcal{Z}(\mathcal{M}))$ such that

$$\int_G u_\alpha(x, h)^2 d\mu(h) = 1, \quad (2)$$

$$\operatorname{ess\,sup}_x \int_G |u_\alpha(g^{-1}x, g^{-1}h) - u_\alpha(x, h)|^2 d\mu(h) \longrightarrow 0 \quad (g \in G). \quad (3)$$

As in the source, “sequence” permits a net; under the usual countability hypotheses one may diagonalize to a literal sequence.

Theorem 1. *Let G be a locally compact Hausdorff group with left Haar measure μ , and let $\theta : G \rightarrow \operatorname{Aut}(\mathcal{M})$ be an amenable action in the sense of (2)–(3). Then there is a net of measurable sets $F_\beta \subset X \times G$ such that, for every β ,*

$$1 \leq \operatorname{ess\,sup}_x \mu((F_\beta)_x) = \operatorname{ess\,inf}_x \mu((F_\beta)_x) < \infty,$$

and the normalized indicators satisfy (3). Hence every amenable action admits a 1-approximating sequence in the sense of Definition 1.3 of [1].

4 Two elementary lemmas

Let $\Delta_G : G \rightarrow (0, \infty)$ be the modular homomorphism and put $H = \ker \Delta_G$. Right translation by any element of H preserves μ .

Lemma 1 (Haar-preserving slots). *If H is noncompact, then for every compact $D \subset G$ and every $n \in \mathbb{N}$ there are $r_0, \dots, r_n \in H$ such that the sets $Dr_0, \dots, Dr_n$ are pairwise disjoint.*

Proof. Suppose $r_0, \dots, r_{k-1}$ have been selected. The intersection

$$H \cap \bigcup_{i < k} D^{-1}Dr_i$$

is compact. Choose r_k outside it, which is possible because H is noncompact. If Dr_k met Dr_i , then $r_k \in D^{-1}Dr_i$, a contradiction. $\square$

Lemma 2 (Equal-volume layer encoding). *Assume H is noncompact. Let $Q \subset G$ be finite and let $p : X \times G \rightarrow [0, \infty)$ be measurable, supported in a common compact set $S \subset G$, with*

$$\int_G p(x, h) d\mu(h) = 1, \quad 0 \leq p(x, h) \leq M.$$

For every sufficiently large N there is a measurable $F \subset X \times G$ whose fibres all have one finite measure $L \geq 1$ and, for every $g \in Q$,

$$\operatorname{ess\,sup}_x \frac{\mu(gF_{g^{-1}x} \triangle F_x)}{L} \leq \operatorname{ess\,sup}_x \|gp_{g^{-1}x} - p_x\|_1 + C_S \frac{M}{N}, \quad (4)$$

where $C_S < \infty$ is independent of N , x , and $g \in Q$. Here $(gp_{g^{-1}x})(h) = p(g^{-1}x, g^{-1}h)$.

Proof. Set $\delta = M/N$ and

$$A_j(x) = \{h : p(x, h) \geq j\delta\}, \quad 1 \leq j \leq N, \quad V(x) = \sum_{j=1}^N \mu(A_j(x)).$$

The functions $x \mapsto \mu(A_j(x))$ are measurable. Pointwise quantization gives

$$0 \leq 1 - \delta V(x) < \delta \mu(S). \quad (5)$$

First suppose G is nondiscrete. Haar measure is nonatomic. Take $L = \delta^{-1}$ and $d(x) = L - V(x)$. By (5), $0 \leq d(x) < \mu(S)$. In a fixed compact measurable set R of measure at least $\mu(S)$, choose a nested measurable family $(R_t)_{0 \leq t \leq \mu(S)}$ with $\mu(R_t) = t$. Such a family follows, for example, by recursive bisection of the nonatomic finite measure space. Put $C_x = R_{d(x)}$.

If G is discrete, normalize Haar measure to counting measure. Put $L = \lceil \delta^{-1} \rceil$. Since $V(x)$ is integral, $d(x) = L - V(x)$ is an integer and (5) implies

$$0 \leq d(x) \leq |S| + 1.$$

Choose a fixed finite ordered set R with at least $|S| + 1$ points and let C_x consist of its first $d(x)$ points. This also makes $(x, h) \mapsto \mathbf{1}_{C_x}(h)$ measurable.

Let $c_S = \mu(S)$ in the nondiscrete case and $c_S = |S| + 1$ in the discrete case. Choose a compact set D containing

$$\bigcup_{g \in Q \cup \{e\}} g(S \cup R).$$

By Lemma 1, choose $r_0, \dots, r_N \in H$ for which the Dr_j are disjoint, and define

$$F_x = C_x r_0 \cup \bigcup_{j=1}^N A_j(x) r_j. \quad (6)$$

All right translations in (6) preserve Haar measure, the pieces are disjoint, and therefore

$$\mu(F_x) = \mu(C_x) + V(x) = d(x) + V(x) = L$$

for every x .

Fix $g \in Q$. The enlarged translated pieces occurring in $gF_{g^{-1}x}$ and F_x still lie in the mutually disjoint Dr_j slots. Consequently

$$\mu(gF_{g^{-1}x} \triangle F_x) = \sum_{j=1}^N \mu(gA_j(g^{-1}x) \triangle A_j(x)) + \mu(gC_{g^{-1}x} \triangle C_x). \quad (7)$$

For $a, b \in [0, M]$,

$$\delta \sum_{j=1}^N |\mathbf{1}_{\{a \geq j\delta\}} - \mathbf{1}_{\{b \geq j\delta\}}| \leq |a - b| + \delta. \quad (8)$$

Apply this at $a = p(g^{-1}x, g^{-1}h)$ and $b = p(x, h)$ and integrate. Since the integrand vanishes off $gS \cup S$,

$$\delta \sum_{j=1}^N \mu(gA_j(g^{-1}x) \triangle A_j(x)) \leq \|gp_{g^{-1}x} - p_x\|_1 + \delta \mu(gS \cup S). \quad (9)$$

The correction term in (7) is at most $2c_S$. In both cases $L \geq \delta^{-1}$, hence $L^{-1} \leq \delta$. Divide (7) by L and use (9) to obtain

$$\frac{\mu(gF_{g^{-1}x} \triangle F_x)}{L} \leq \|gp_{g^{-1}x} - p_x\|_1 + \delta(\mu(gS \cup S) + 2c_S).$$

Left invariance gives $\mu(gS \cup S) \leq 2\mu(S)$, proving (4) with $C_S = 2\mu(S) + 2c_S$. $\square$

5 Proof of the theorem

Proof of Theorem 1. We divide into two cases.

Case 1: G is amenable. The Følner–Reiter criterion gives, for every finite $Q \subset G$ and $\varepsilon > 0$, a relatively compact measurable set A of positive finite measure such that

$$\frac{\mu(gA \triangle A)}{\mu(A)} < \varepsilon \quad (g \in Q).$$

For completeness, this follows by choosing $f \in C_c(G)_+$ with $\int f = 1$ and $\sum_{g \in Q} \|gf - f\|_1 < \varepsilon$, applying layer cake to $A_t = \{f > t\}$, and selecting one level at which the ratio of the summed boundaries to $\mu(A_t)$ is small. In a noncompact group, A can be replaced by a right translate or a disjoint union of right translates so that its measure is at least one without changing these ratios. Indeed, if $\ker \Delta_G$ is noncompact, use finitely many disjoint right translates from that kernel. If $\ker \Delta_G$ is compact, then noncompactness of G forces the right-Haar scaling factors to be unbounded, and one right translate suffices. In a compact group take $A = G$ with the standard probability

normalization of Haar measure. Put $F_x = A$ for every x . Then (1) tends to zero, independently of the action, and all fibres have the same measure.

Case 2: G is nonamenable. The quotient G/H is abelian, hence amenable. Amenability is stable under extensions [2, Chapter 2]; thus H cannot be amenable. In particular H is noncompact.

Fix a finite $Q \subset G$ and $\varepsilon > 0$. By amenability of the action, choose one positive compactly supported approximant u satisfying (2) and

$$\operatorname{ess\,sup}_x \|gu_{g^{-1}x} - u_x\|_2 < \varepsilon/4 \quad (g \in Q). \quad (10)$$

Put $p_x = u_x^2$. It has integral one, a common compact support S , and a finite uniform bound M . Cauchy–Schwarz and (2) give

$$\begin{aligned} \|gp_{g^{-1}x} - p_x\|_1 &= \int_G |(gu_{g^{-1}x})^2 - u_x^2| d\mu \\ &\leq \|gu_{g^{-1}x} - u_x\|_2 \|gu_{g^{-1}x} + u_x\|_2 \leq 2\|gu_{g^{-1}x} - u_x\|_2. \end{aligned} \quad (11)$$

Choose N so large that the second term on the right of (4) is less than $\varepsilon/2$ and that the resulting common fibre measure is at least one. Lemma 2, (10), and (11) now yield

$$\operatorname{ess\,sup}_x \frac{\mu(gF_{g^{-1}x} \triangle F_x)}{L} < \varepsilon \quad (g \in Q).$$

By (1), the normalized indicators have squared L_2 defect less than ε on Q . Direct the construction by the pairs (Q, ε) , ordered by increasing Q and decreasing ε . This produces the required 1-approximating net. $\square$

6 Verification report

The proof was audited through the following independent checkpoints.

1. **Modular dichotomy.** If G is nonamenable, then $\ker \Delta_G$ is nonamenable and hence noncompact, because the quotient is abelian and amenability is extension-stable.
2. **Slot separation.** At the k th choice only the compact set $H \cap \bigcup_{i < k} D^{-1}Dr_i$ is forbidden. This cannot exhaust noncompact H .
3. **Exact volume.** The level sum has deficit at most $\mu(S)$ (or at most $|S| + 1$ in the discrete case). The correction layer fills that deficit exactly, and every slot preserves Haar measure.
4. **Defect estimate.** Equation (8) is the floor-function bound $\delta|[a/\delta] - [b/\delta]| \leq |a - b| + \delta$. The arbitrary correction costs at most $2c_S/L$.
5. **Measurability.** Tonelli gives measurability of all level volumes. A nonatomic finite measure space admits a nested family of every intermediate measure; composing that family with the measurable deficit gives a measurable correction field. The discrete selection is finite-valued.
6. **Finite regression check.** Running `python code/verify_quantized_layers.py` tests 50,000 random scalar floor inequalities and 5,000 random discrete volume corrections. It prints **PASS**. This check is not used as proof.

7 Scope, limitations, and novelty search

The result settles the $C = 1$ conjecture as stated in the source paper, using the source paper’s own approximating-net characterization of an amenable action. It does not address the distinct conjecture later in that paper about whether an intertwining complete isometry characterizes amenability.

The numerical lower bound “ $1 \leq \mu(F_x)$ ” depends on the arbitrary scalar normalization of Haar measure. The construction makes L arbitrarily large in the nonamenable case; the standard Følner amplification does the same for noncompact amenable groups, while compact Haar measure is conventionally normalized to one.

A bounded novelty check was performed on 12–13 August 2026. It searched the run’s registry, solution, attempt, and proof-gap indexes; exact arXiv id and paper-title queries; and the phrases “every amenable action admits a 1-approximating sequence”, “ C -approximating sequences”, and “accretivity constant” together with amenable action/crossed product. The search also inspected the citation neighborhood containing arXiv:2009.03019 and later multiplier-transference work. No paper claiming this conjecture was found. A 2026 thesis by the source author still records examples with $C = 1$ rather than a general theorem. This is not an exhaustive MathSciNet/zblMATH citation review, so mathematical confidence is high but novelty confidence is moderate pending expert bibliographic review.

8 Human-review recommendation

Review as a candidate full solution. The most valuable checks are: (i) the measurable nested correction family in the exact measure-theoretic framework intended by Definition 1.3; (ii) the harmless net/sequence convention; and (iii) the standard extension theorem used to force noncompactness of $\ker \Delta_G$. No computational or unproved mathematical dependency remains.

References

- [1] A. M. González-Pérez, *Crossed-Product Extensions of L_p -Bounds for Amenable Actions*, J. Funct. Anal. 274 (2018), no. 10, 2846–2883; [arXiv:1611.08486](#).
- [2] A. L. T. Paterson, *Amenability*, Mathematical Surveys and Monographs 29, American Mathematical Society, 1988.